

Particle Filters for Inference in Continuous State Space

[TODO: Author name]

[TODO: Date]

Abstract

[TODO: Abstract: the question, what was compared, the main findings in two or three sentences.]

1 Introduction

[TODO: Introduction.]

2 SLAM as a Graphical Model

2.1 Dynamic Bayesian network

2.2 Rao-Blackwellization

2.3 From discrete to continuous inference

[TODO: Background.]

3 Particle Filters as Approximate Inference

3.1 Sequential importance sampling

3.2 Proposal distributions

3.3 Resampling

3.4 MCMC moves

[TODO: Particle filters.]

4 Method

4.1 Filters and worlds

4.2 Implementation

4.3 Metrics

[TODO: Method.]

5 Experiments and Results

5.1 Resampling scheme

[TODO: E01: question, setup and findings.]

[TODO: Table for E01_resampling_scheme_localization: run the experiment first.]

[TODO: Figure metrics for E01_resampling_scheme_localization: run the experiment first.]

[TODO: Table for E01_resampling_scheme_slam: run the experiment first.]

[TODO: Figure metrics for E01_resampling_scheme_slam: run the experiment first.]

5.2 When to resample

[TODO: E02: question, setup and findings.]

[TODO: Table for E02_when_to_resample: run the experiment first.]

[TODO: Figure metrics for E02_when_to_resample: run the experiment first.]

[TODO: Figure trace_ess for E02_when_to_resample: run the experiment first.]

5.3 Proposal distribution

[TODO: E03: question, setup and findings.]

[TODO: Table for E03_proposal_localization: run the experiment first.]

[TODO: Figure metrics for E03_proposal_localization: run the experiment first.]

[TODO: Table for E03_proposal_slam: run the experiment first.]

[TODO: Figure metrics for E03_proposal_slam: run the experiment first.]

5.4 Number of particles

[TODO: E04: question, setup and findings.]

[TODO: Table for E04_particle_count: run the experiment first.]

[TODO: Figure metrics for E04_particle_count: run the experiment first.]

5.5 Resample-move

[TODO: E05: question, setup and findings.]

[TODO: Table for E05_resample_move: run the experiment first.]

[TODO: Figure metrics for E05_resample_move: run the experiment first.]

5.6 Real data: gmapping

[TODO: E06 (optional): question, setup and findings. Remove this subsection if E06 is not run.]

[TODO: Table for E06_gmapping_resampling: run the experiment first.]

[TODO: Figure metrics for E06_gmapping_resampling: run the experiment first.]

6 Discussion

[TODO: Discussion.]

7 Conclusion

[TODO: Conclusion.]

A Upstream Code and Corrections

[TODO: Appendix: upstream code and corrections.]